

All athletes at the Olympic games are tested for performance-enhancing steroid drug use. The imperfect test gives positive results (indicating drug use) for 90% of all “dirty” athletes who used steroids but also (and incorrectly) for 2% of “clean” athletes who do not use steroids. Suppose that 5% of all registered athletes use steroids. If an athlete is tested negative, what is the probability that he/she uses steroids?

*SOLUTION.* Let  $S = \{\text{steroid user}\}$  and  $N = \{\text{test is negative}\}$ .

It is given that  $\mathbf{P}\{S\} = 0.05$ ,  $\mathbf{P}\{\bar{N}|S\} = 0.9$ , and  $\mathbf{P}\{\bar{N}|\bar{S}\} = 0.02$ .

By the complement rule,  $\mathbf{P}\{\bar{S}\} = 0.95$ ,  $\mathbf{P}\{N|S\} = 0.1$ , and  $\mathbf{P}\{N|\bar{S}\} = 0.98$ .

By the Bayes Rule,

$$\begin{aligned}\mathbf{P}\{S|N\} &= \frac{\mathbf{P}\{N|S\} \mathbf{P}\{S\}}{\mathbf{P}\{N|S\} \mathbf{P}\{S\} + \mathbf{P}\{N|\bar{S}\} \mathbf{P}\{\bar{S}\}} \\ &= \frac{(0.1)(0.05)}{(0.1)(0.05) + (0.98)(0.95)} = \boxed{5/936 \text{ or } 0.00534}\end{aligned}$$